HUGGING FACE · STANFORD

SmolVLM: Redefining Small and Efficient Multimodal Models

Compact Vision-Language Models for Edge and Mobile Deployment

"Sub-1GB inference · 13 benchmarks · Fully open-source"

April 2026

arXiv: 2504.05299

MOTIVATION

Large VLMs Are Powerful but Impractical for Edge Deployment

- **State-of-the-art VLMs require 20–80 GB GPU RAM**
  GPT-4V, LLaVA-1.6 — unfeasible on mobile or edge hardware
- **Best open-source baseline: Molmo-A1B-7B needs 27.7 GB**
  Even "efficient" models remain impractical for on-device use
- **Edge devices budget: 2–16 GB RAM**
  Smartphones, embedded MCUs, IoT — memory-constrained by design
- **Cloud dependency introduces latency and privacy risks**
  Connectivity required; user data leaves device

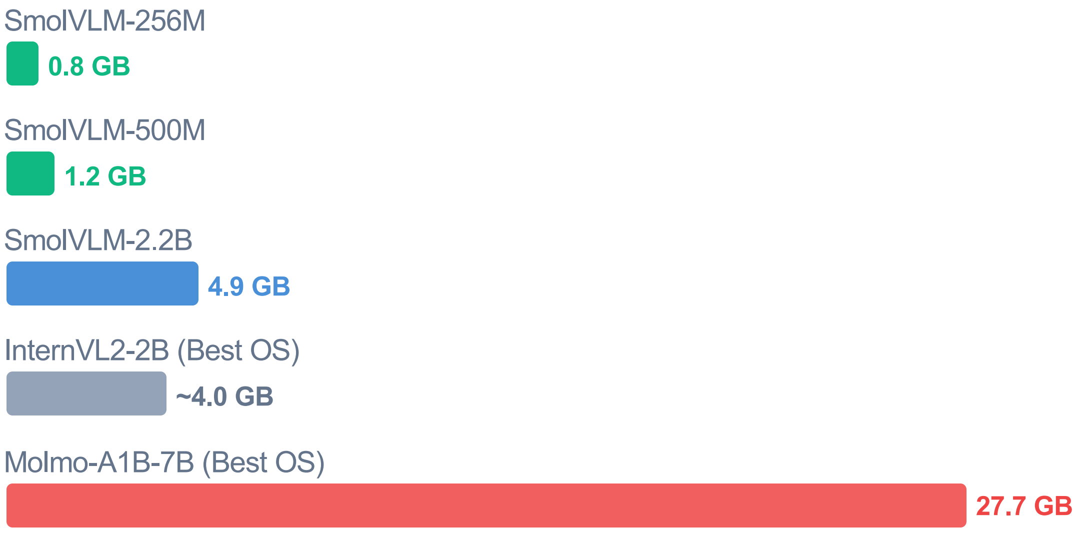
THE CHALLENGE

Achieve competitive multimodal performance under sub-1 GB RAM

SMOLVLM'S ANSWER

3 sizes: 256M, 500M, 2.2B — all open-source, edge-ready

GPU RAM Usage Comparison (Batch = 1)



Bar width proportional to RAM usage

34× less RAM than Molmo-A1B-7B
SmolVLM-256M (0.8 GB) vs. Molmo-A1B-7B (27.7 GB)
SmolVLM-256M beats 300× larger Idefics-80B

ARCHITECTURE

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

A five-stage pipeline converts raw images into language model responses with aggressive efficiency at every step.

01 Input Image
Split into tiles
+ global view
sub-images

High-res tiling

02 Vision Encoder
SigLIP-B/16
93M params
Encodes each tile

Compact encoder

03 Pixel Shuffle
Space-to-depth
r=4 → 16× fewer
visual tokens

Token compression

04 Linear Projection
Project visual
tokens into
SmolLM2 space

Embedding bridge

05 SmolLM2
Language Model
135M – 2.2B params
8K–16K context

Text generation

High-res tiles
+ thumbnail

93M SigLIP-B/16
Optimal for small LMs

r=4: 16× reduction
Key for ≤500M models

Linear layer
Modality bridging

SmolLM2 backbone
Concat text + visual

VIDEO: Frames are sampled and processed through the identical pipeline — no separate video encoder required
Concatenated visual tokens across frames feed directly into SmolLM2 for temporal understanding

Three architectural findings drive the design:

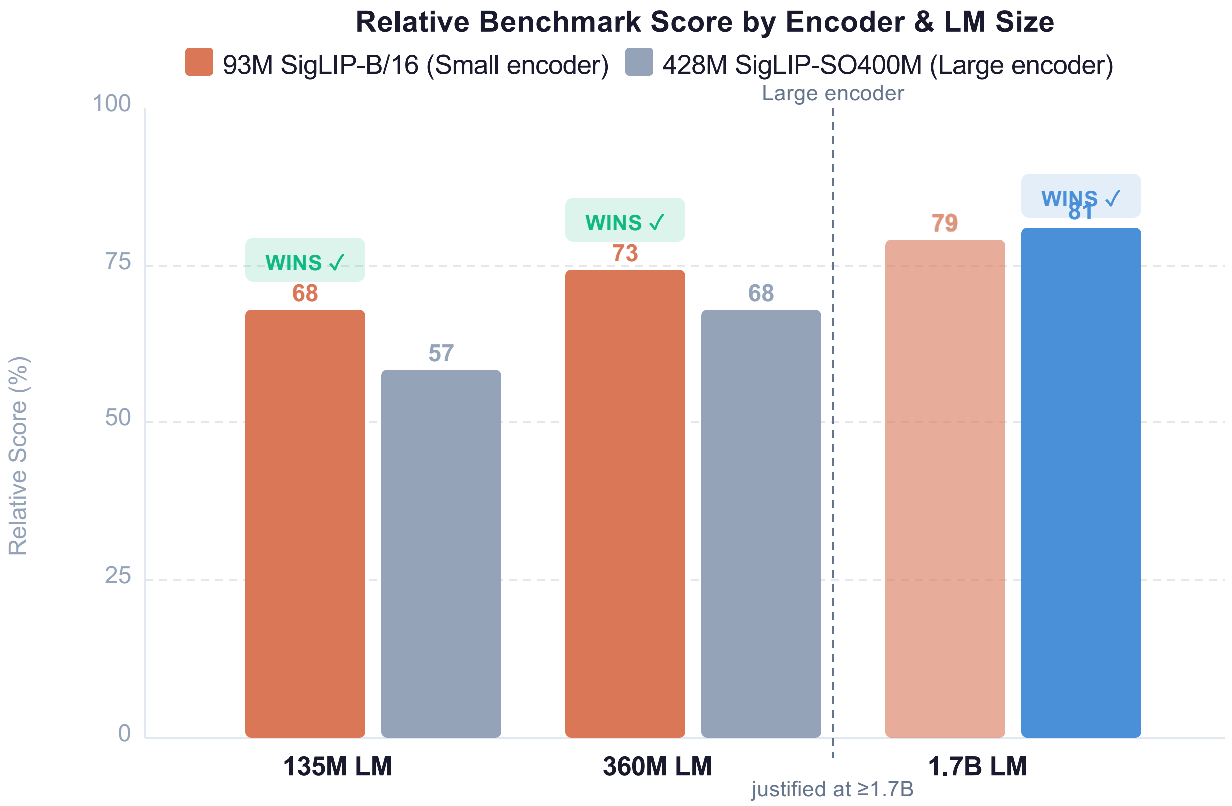
- Finding 1: Balanced encoder-LM compute allocation
- Finding 2: Extended context (2K → 16K tokens)
- Finding 3: Aggressive token compression (r=4)

Source: SmolVLM (arXiv:2504.05299) · Figure 3

FINDING 1 — COMPUTE ALLOCATION

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

KEY INSIGHT 93M SigLIP-B/16 outperforms 428M SigLIP-SO400M for small LMs — 4.6× fewer vision params, better results



Design Rule: Match Encoder to LM Capacity

93M SigLIP-B/16 wins at small scale
Outperforms 428M SigLIP-SO400M paired with 135M and 360M LMs

Large encoder justified at ≥1.7B LM scale
10% additional parameter overhead becomes worthwhile only at this LM scale

Bigger vision encoder ≠ better accuracy
For compact VLMs, balanced allocation matters more than raw encoder capacity

SMOLVLM DESIGN DECISION
256M & 500M variants → 93M SigLIP-B/16
4.6× fewer vision encoder parameters vs. SO400M
Lower compute cost, higher per-parameter efficiency

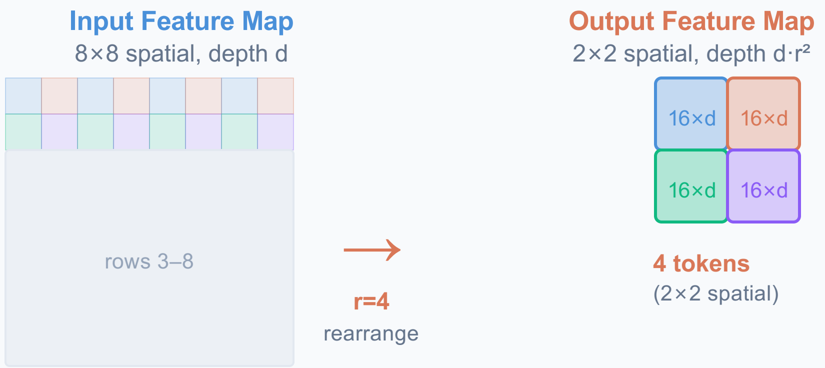
428M SigLIP-SO400M vs **93M** SigLIP-B/16 ✓

FINDING 3 — TOKEN COMPRESSION

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

KEY INSIGHT r=4 reduces 16× more visual tokens than no compression — ideal for models ≤500M where attention cost dominates

Pixel Shuffle (Space-to-Depth) Mechanism



64 tokens
(8×8 spatial)

16× Fewer Visual Tokens

64 tokens → 4 tokens (r=4) · Channel depth expands 16× (d → 16d)

Token Reduction = r^2

Spatial ($H \times W$) → ($H/r \times W/r$) · Channels (d) → ($d \times r^2$)

Benefits for small VLMs:

- Reduced attention overhead (attention scales $O(n^2)$ with tokens)
- Enables processing more image tiles within 8K/16K context budgets
- Faster inference — critical for mobile and edge deployment

Shuffle Factor Trade-off by Model Scale

Factor	Reduction	Best For
r = 1	1×	Very large models
r = 2	4×	≥ 1.7B LM scale
r = 4 ✓ SmolVLM	**16×**	**≤ 500M LM scale** 256M & 500M ✓

Why r=4 works for small models

Attention is the bottleneck
Small LMs have limited context capacity; fewer tokens means each fits in 8K–16K window

Large models retain more spatial detail
≥1.7B LMs can afford r=2 (4× reduction) to preserve spatial features for complex tasks

RESULT
SmolVLM-256M/500M use r=4: 16× token reduction with minimal accuracy loss on all 13 benchmarks

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

KEY INSIGHT Increasing context from 2K to 8K–16K tokens enables more image tiles, higher effective resolution, and better performance

Standard short-context VLMs: 2K token limit

With r=4 compression, each image tile produces ~64 visual tokens
A 2K context handles only a few tiles → low effective resolution
Performance significantly limited at 2K context

RoPE base increased: 10K → 273K

Rotary Position Embedding base scaled 27× to support long positional sequences without accuracy degradation
Enables robust positional encoding up to 16K tokens

Fine-tuned on mixed long/short-context data

Training data includes both long-context (multi-tile images, video sequences) and short-context samples
Maintains performance on both short and long-context tasks

PERFORMANCE IMPACT

Significant benchmark improvement: 2K → 8K/16K context extension
Multi-image reasoning and video understanding benefit most

Context budget (with r=4, ~64 tokens/tile)

2K → ~31 tiles · 8K → ~125 tiles · 16K → ~250 tiles

Context Configuration by Model Size

Model	Context	RoPE Base
SmolVLM-256M	8K tokens	273K
SmolVLM-500M	8K tokens	273K
SmolVLM-2.2B flagship	16K tokens	273K
Baseline (all) before extension	2K tokens	10K

What is RoPE base scaling?

Rotary Position Embeddings encode token positions. A higher base value (10K → 273K) allows accurate encoding of longer sequences without needing architectural changes to the model.

WHY THIS MATTERS FOR VISION

More context tokens = more image tiles fit in a single forward pass
Higher-resolution images, multi-image QA, and video frames all benefit directly from expanded context without additional architectural overhead

BENCHMARKS

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

SmolVLM-256M
44.0% avg (13 benchmarks)
RAM: 0.8 GB · batch=1 · sub-1GB

SmolVLM-500M
51.0% avg (13 benchmarks)
RAM: 1.2 GB

SmolVLM-2.2B
59.8% avg (13 benchmarks)
RAM: 4.9 GB

Molmo-A1B-7B
(Best OS baseline)
27.7 GB RAM

Category	Benchmark	256M	500M	2.2B	Best OS Baseline		Baseline RAM
SINGLE IMAGE	OCRBench	52.6%	61.0%	**72.9%**	54.7% (Molmo-A1B)		27.7 GB
	AI2D	46.4%	59.2%	**70.0%**	71.0% (Molmo-A1B)		27.7 GB
	ChartQA	55.6%	62.8%	**68.7%**	48.0% (Molmo-A1B)	↑ **SmolVLM wins**	27.7 GB
	TextVQA	50.2%	60.2%	**73.0%**	61.5% (Molmo-A1B)		27.7 GB
	DocVQA	58.3%	70.5%	**80.0%**	77.7% (Molmo-A1B)	↑ **SmolVLM wins**	27.7 GB
	ScienceQA	73.8%	80.0%	**89.6%**	87.5% (Molmo-A1B)		27.7 GB
MULTI-TASK	MMMU	29.0%	33.7%	**42.0%**	33.9% (Molmo-A1B)		27.7 GB
	MathVista	35.9%	40.1%	**51.5%**	37.6% (Molmo-A1B)		27.7 GB
	MMStar	34.6%	38.3%	**46.0%**	43.1% (Molmo-A1B)		27.7 GB
VIDEO	Video-MME	33.7%	42.2%	**52.1%**	45.0% (InternVL2-2B)		~4.0 GB
	MLVU	40.6%	47.3%	**55.2%**	48.2% (InternVL2-2B)		~4.0 GB
	AVERAGE (All 13)	**44.0%**	**51.0%**	**59.8%**	—		**vs 27.7 GB**

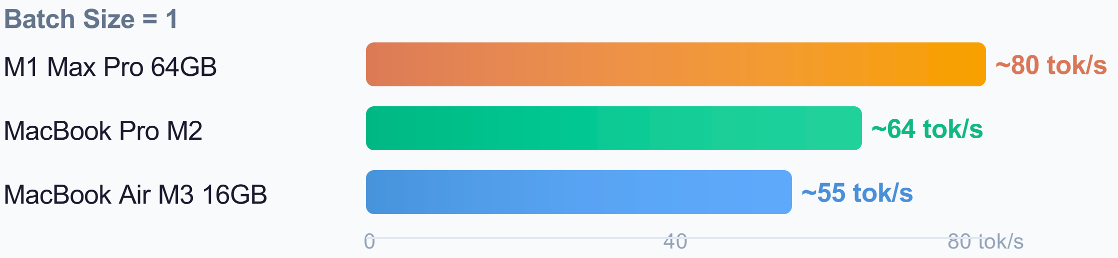
KEY FINDING: SmolVLM-256M (0.8 GB) outperforms the 300× larger Idefics-80B · SmolVLM-2.2B rivals models using twice the GPU memory
13 benchmarks: Single-Image (6) · Multi-task (3) · Video (2)

Source: SmolVLM (arXiv:2504.05299) · Molmo-A1B-7B baseline RAM: 27.7 GB · InternVL2-2B used for video benchmarks

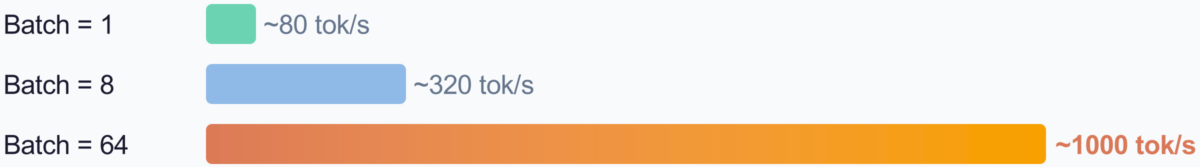
Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

KEY INSIGHT SmolVLM-256M achieves real-time multimodal inference on consumer Apple Silicon — entirely on-device, no cloud

SmolVLM-256M Throughput (tokens/second)



Batch Size Scaling (256M, high-end device)



Throughput scales ~linearly with batch size for the 256M model

0.8 GB
RAM at batch=1

256M
model parameters

HuggingSnap: On-Device Multimodal AI

HuggingSnap iOS App
Processes photos and video entirely on-device
No cloud dependency · Full privacy preservation

On-Device Deployment Benefits

1. **Privacy-preserving**
   No user data leaves the device — GDPR-friendly
2. **Zero-latency inference**
   No network round-trip — real-time response
3. **Works offline**
   No internet required — functional in any environment
4. **Accessible to everyone**
   Smartphones, tablets, laptops — consumer hardware

Fully Open-Source Release
All weights, training data, code, and HuggingSnap app
published for reproducibility and on-device AI research

Target platforms: iPhones / iPads · Apple Silicon · Edge MCUs / IoT

Source: SmolVLM (arXiv:2504.05299) · Figure 2 · HuggingSnap iOS app · Apple Silicon throughput benchmarks

KEY TAKEAWAYS

Design Principles for Efficient Multimodal Models

1 Balance Encoder-LM Compute

Match vision encoder size to LM capacity. 93M SigLIP-B/16 outperforms 428M SigLIP-SO400M for models ≤500M params.

Bigger encoder ≠ better accuracy at small LM scale

Finding 1

2 Extend Context Length

Push RoPE base to 273K; extend context from 2K to 8K–16K tokens for higher-resolution multi-tile image processing.

More context = more tiles = better vision coverage

Finding 2

3 Compress Tokens Aggressively

Use pixel shuffle r=4 for models ≤500M — 16× token reduction with minimal accuracy loss.

Eases attention, improves long-context modeling

Finding 3

4 Sub-1GB Inference Is Achievable

SmolVLM-256M (0.8 GB) outperforms the 300× larger Idefics-80B model.

Efficiency > raw scale for on-device AI

SmolVLM-256M

5 Open-Source for Reproducibility

Weights, training data, code, and HuggingSnap mobile app released.

Promotes on-device AI research worldwide

Hugging Face · Stanford

SmolVLM demonstrates that architectural efficiency — not raw scale — is the key to practical on-device AI

Balanced compute allocation + aggressive token compression + extended context = competitive VLMs at a fraction of the memory

0.8 GB	13	16K	300×
min RAM (256M)	benchmarks evaluated	max context tokens	larger Idefics-80B beaten